4 The beta function

4.1 Introduction

A particular combination of gamma functions is given a name because it has a simple and useful integral representation. The beta function, or Eulerian integral of the first kind, is

$$B(x, y) = \int_0^1 u^{x-1}(1-u)^{y-1}\,du, \tag{1}$$

where x and y have positive real parts. Since the integral will be shown to have the value $\Gamma(x)\Gamma(y)/\Gamma(x+y)$, it can be used for alternative proofs of some properties of the gamma function. With various changes of integration variable it is seen frequently in applied mathematics, for example in the theory of high-energy particle physics.

If x and y are real and positive, the integrand of (1) occurs in statistics as the frequency function of the beta distribution. To normalize the distribution to unit total probability, we divide (1) by $B(x, y)$. Complex values of x and y hold little interest for statistics, but the integrand still defines a complex measure on the unit interval which can be normalized as done previously:

$$d\mu(u) = \frac{u^{x-1}(1-u)^{y-1}}{B(x, y)}\,du, \tag{2}$$

$$\int_0^1 d\mu(u) = 1, \qquad x, y \in \mathbb{C}_>. \tag{3}$$

The beta distribution for more than one random variable can likewise be extended to complex values of the parameters. For example we may define the complex measure

$$d\mu(u, v) = \frac{u^{x-1}v^{y-1}(1 - u - v)^{z-1}}{B(x, y, z)} \, du \, dv, \tag{4}$$

$$\int d\mu(u, v) = 1, \qquad x, y, z \in \mathbb{C}_>, \tag{5}$$

where the triangular region of integration is $\{(u, v) \colon u \geq 0, v \geq 0, u + v \leq 1\}$. Equations (4) and (5) determine $\mathrm{B}(x, y, z)$, which we shall call the beta function of three variables.

Because Dirichlet (1839) evaluated an integral of this type in several variables, we shall call the complex measure (4) or its analog in any number of variables a Dirichlet measure. In later chapters Dirichlet measures will be fundamental in defining special functions.

4.2 The beta function of two variables

To avoid restricting the real parts of the variables, we define the beta function in terms of gamma functions (but see Exercise 4.3-3).

DEFINITION 4.2-1 Let $x, y \in U$. The beta function is defined by

$$\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x + y)}. \tag{1}$$

REPRESENTATION 4.2-2 If $x, y \in \mathbb{C}_>$, then

$$\mathrm{B}(x, y) = \int_0^1 u^{x-1}(1 - u)^{y-1} \, du, \tag{2}$$

$$\mathrm{B}(x, y) = 2\int_0^{\pi/2} (\cos\theta)^{2x-1}(\sin\theta)^{2y-1} \, d\theta, \tag{3}$$

$$\mathrm{B}(x, y) = \int_0^\infty \frac{t^{x-1}}{(t + 1)^{x+y}} \, dt. \tag{4}$$

Proof It is sufficient to extend slightly the second proof of Corollary 3.4-2. We consider first the case in which $x, y \in \mathbb{R}_>$. Using Cartesian coordinates ξ, η and plane polar coordinates r, θ, we have

$$\int_0^\infty \int_0^\infty \xi^{2x-1}\eta^{2y-1}e^{-\xi^2-\eta^2} \, d\xi \, d\eta$$
$$= \int_0^{\pi/2} \int_0^\infty r^{2x+2y-1}(\cos\theta)^{2x-1}(\sin\theta)^{2y-1}e^{-r^2} \, dr \, d\theta.$$

Since the integrands are positive, the integrals may be evaluated as repeated integrals by Fubini's theorem (Appendix B.2). Then

$$\int_0^\infty \xi^{2x-1} e^{-\xi^2}\, d\xi \int_0^\infty \eta^{2y-1} e^{-\eta^2}\, d\eta$$
$$= \int_0^\infty r^{2x+2y-1} e^{-r^2}\, dr \int_0^{\pi/2} (\cos\theta)^{2x-1} (\sin\theta)^{2y-1}\, d\theta. \quad (5)$$

Setting $\xi = t^{1/2}$ yields

$$\int_0^\infty \xi^{2x-1} e^{-\xi^2}\, d\xi = \frac{1}{2}\int_0^\infty t^{x-1} e^{-t}\, dt = \frac{1}{2}\Gamma(x).$$

Thus

$$\tfrac{1}{4}\Gamma(x)\Gamma(y) = \tfrac{1}{2}\Gamma(x+y)\int_0^{\pi/2} (\cos\theta)^{2x-1} (\sin\theta)^{2y-1}\, d\theta,$$

which is (3). The substitutions $\cos\theta = u^{1/2}$ and $u = t/(t+1)$ then lead to (2) and (4).

If $x, y \in \mathbb{C}_>$, all integrands can be replaced by their absolute values by simply replacing x and y by $\operatorname{Re} x$ and $\operatorname{Re} y$. After replacement, the iterated integrals in (5) are still finite, as we can see from the real case. By Fubini's theorem, (5) is then valid also when x and y are complex. □

The symmetry property

$$B(x, y) = B(y, x) \quad (6)$$

is obvious from (1). For the integral representations it means that there is no essential distinction between the two ends of the interval of integration. In (2), for example, x has the same relation to the endpoint $u = 0$ as y has to the endpoint $u = 1$. Replacement of u by $1 - u$ as the integration variable reverses the direction of integration and associates y with the endpoint $u = 0$ and x with the endpoint $u = 1$, the value of the integral being unchanged. Symmetries of a similar kind will be important in subsequent sections.

4.3 The beta function of several variables

The definition and integral representation of the beta function will now be extended to three or more variables.

DEFINITION 4.3-1 Let $b = (b_1, \ldots, b_k) \in U^k$, $k \geq 2$. The beta function of k variables is defined by

$$B(b) = B(b_1, \ldots, b_k) = \frac{\Gamma(b_1)\cdots\Gamma(b_k)}{\Gamma(b_1 + \cdots + b_k)}. \quad (1)$$

For $k = 2$ the ordinary beta function is represented by an integral (4.2-2) over the unit interval $0 \le u \le 1$. If $k = 3$, we integrate over a triangular region in the (u, v) plane with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. This is the set $\{(u, v)\colon u \ge 0,\ v \ge 0,\ u + v \le 1\}$ and will be called the standard simplex in $\mathbb{R}^2$ (see Fig. 4.3-1). The points (u, v) of the simplex are in one-to-one

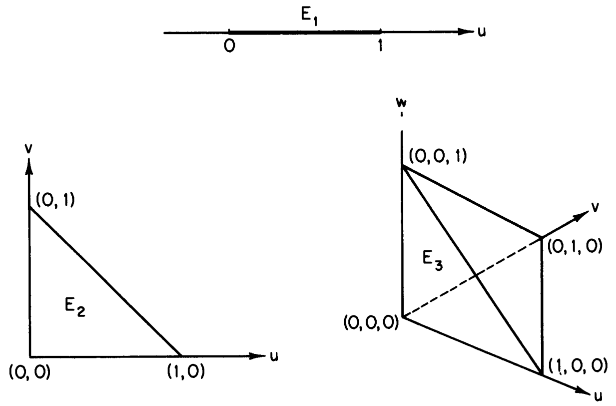

Figure 4.3-1 The standard simplex E_n.

correspondence with the triples $(u, v, 1 - u - v)$ of nonnegative weights with unit sum. The standard simplex in $\mathbb{R}^3$ is $\{(u, v, w)\colon u \ge 0,\ v \ge 0,\ w \ge 0,\ u + v + w \le 1\}$, a solid tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. In general we denote the standard simplex in $\mathbb{R}^n$, $n \ge 1$, by

$$E = E_n = \{(u_1, \ldots, u_n)\colon u_1 \ge 0, \ldots, u_n \ge 0,\ u_1 + \cdots + u_n \le 1\}. \tag{2}$$

The points $(u_1, \ldots, u_n)$ of the simplex are in one-to-one correspondence with the $(n + 1)$-tuples $(u_1, \ldots, u_n, 1 - u_1 - \cdots - u_n)$ of nonnegative weights with unit sum. Unless the value of n needs attention, we shall write E in place of E_n. The interior of E is denoted by

$$\operatorname{int}(E) = \{(u_1, \ldots, u_n)\colon u_1 > 0, \ldots, u_n > 0,\ u_1 + \cdots + u_n < 1\}. \tag{3}$$

REPRESENTATION 4.3-2 Let $b \in \mathbb{C}_>{}^k$, $k \ge 2$, and let $E = E_{k-1}$ be the standard simplex in $\mathbb{R}^{k-1}$. Then

$$\mathrm{B}(b_1, \ldots, b_k) = \int_E u_1^{b_1 - 1} \cdots u_{k-1}^{b_{k-1} - 1}(1 - u_1 - \cdots - u_{k-1})^{b_k - 1}\, du_1 \cdots du_{k-1}. \tag{4}$$

Proof Denote the integral by $I_k(b_1, \ldots, b_k)$ and note that $I_2(b_1, b_2) = \mathrm{B}(b_1, b_2)$ by (4.2-2). For $k \ge 3$ we consider first the case $b \in \mathbb{R}_>{}^k$. Then the integrand is positive, and the integral may be evaluated as an iterated integral by Fubini's theorem (see Appendix B.2). Integrating first over u_{k-1} and defining $v = 1 - u_1 - \cdots - u_{k-2}$, we find

$$I_k(b_1, \ldots, b_k) = \int u_1^{b_1 - 1} \cdots u_{k-2}^{b_{k-2}-1}\, du_1 \cdots du_{k-2} \times \int_0^v u_{k-1}^{b_{k-1}-1}(v - u_{k-1})^{b_k - 1}\, du_{k-1}. \quad (5)$$

The region of integration for the outer integral is

$$\{(u_1, \ldots, u_{k-2}) : u_1 \geq 0, \ldots, u_{k-2} \geq 0,\, 1 - u_1 - \cdots - u_{k-2} \geq 0\} = E_{k-2}.$$

If we set $u_{k-1} = xv$, the inner integral becomes

$$v^{b_{k-1}+b_k-1} \int_0^1 x^{b_{k-1}-1}(1-x)^{b_k-1}\, dx = v^{b_{k-1}+b_k-1}\mathrm{B}(b_{k-1}, b_k)$$

by (4.2-2). Thus

$$\begin{aligned} I_k(b_1, \ldots, b_k) &= \mathrm{B}(b_{k-1}, b_k) \int_{E_{k-2}} u_1^{b_1-1} \cdots u_{k-2}^{b_{k-2}-1} \\ &\quad \times (1 - u_1 - \cdots - u_{k-2})^{b_{k-1}+b_k-1}\, du_1 \cdots du_{k-2} \\ &= I_{k-1}(b_1, \ldots, b_{k-2}, b_{k-1} + b_k)\mathrm{B}(b_{k-1}, b_k). \end{aligned} \quad (6)$$

We now make the inductive assumption that I_{k-1} is the beta function of $k-1$ variables, which is true for I_2. By (6) and (1),

$$I_k(b_1, \ldots, b_k) = \mathrm{B}(b_1, \ldots, b_{k-2}, b_{k-1} + b_k)\mathrm{B}(b_{k-1}, b_k) = \mathrm{B}(b_1, \ldots, b_k).$$

Hence (4) follows by induction for $k \geq 3$ if $b \in \mathbb{R}^k$.

The same proof holds for the complex case if we can establish (5) when $b \in \mathbb{C}_>{}^k$. The integrand of (5) can be replaced by its absolute value simply by replacing each b_i by $\operatorname{Re} b_i$. The iterated integral of the absolute value is finite by the preceding proof for the real case. Then Fubini's theorem ensures that (5) is valid in the complex case. □

According to (1), $\mathrm{B}(b_1, \ldots, b_k)$ is symmetric in $b_1, \ldots, b_k$, and it is worthwhile to verify directly that the integral on the right side of (4) has the same property. Symmetry in $b_1, \ldots, b_{k-1}$ is made evident by simply relabeling the variables of integration, so it suffices to prove symmetry in b_1 and b_k. We define

$$u_k = 1 - u_1 - u_2 - \cdots - u_{k-1}, \quad (7)$$

and change variables from $u_1, \ldots, u_{k-1}$ to $u_2, \ldots, u_k$. The Jacobian determinant of the transformation has unit magnitude, and the region of integration changes from

$$\{(u_1, \ldots, u_{k-1}) : u_1 \geq 0, \ldots, u_{k-1} \geq 0,\, u_1 + \cdots + u_{k-1} \leq 1\} \quad (8)$$

to

$$\{(u_2, \ldots, u_k) : u_2 \geq 0, \ldots, u_k \geq 0, u_2 + \cdots + u_k \leq 1\}. \tag{9}$$

The last two inequalities in (9) result from the last and first in (8). Thus the region of integration is again E in terms of the new variables, and the integral on the right side of (4) becomes

$$\int_E (1 - u_2 - \cdots - u_k)^{b_1 - 1} u_2^{b_2 - 1} \cdots u_k^{b_k - 1} \, du_2 \cdots du_k. \tag{10}$$

Relabeling u_k as u_1 restores this to the original form with b_1 and b_k interchanged. Hence the original integral is indeed symmetric in b_1 and b_k.

We may think of the variables $b_1, \ldots, b_k$ as being attached to the vertices of the simplex $E = E_{k-1}$ as follows: b_1 to the vertex $(1, 0, \ldots, 0)$, b_2 to $(0, 1, 0, \ldots, 0), \ldots, b_{k-1}$ to $(0, \ldots, 0, 1)$, and finally b_k to $(0, \ldots, 0)$. We have verified that the vertex at the origin is on an equal footing with all the others. The points $(u_1, \ldots, u_{k-1})$ of E_{k-1} are in one-to-one correspondence with the k-tuples $(u_1, \ldots, u_{k-1}, 1 - u_1 - \cdots - u_{k-1})$ of nonnegative weights with unit sum, and the last of these weights is the quantity u_k defined in (7), which has the value unity at the vertex $(0, \ldots, 0)$. The full symmetry in $1, \ldots, k$ can be exhibited by writing the $(k-1)$-fold integral in (4) as a k-fold integral with a Dirac delta function in the integrand:

$$\mathrm{B}(b) = \int_0^1 \cdots \int_0^1 u_1^{b_1 - 1} \cdots u_k^{b_k - 1} \, \delta(1 - u_1 - \cdots - u_k) \, du_1 \cdots du_k. \tag{11}$$

4.4 Dirichlet measures

The integral representation of the beta function fixes the normalizing constant of a measure defined on the standard simplex in $\mathbb{R}^{k-1}$. The symmetry discussed at the end of the last section implies that integration with respect to this measure conserves permutation symmetry of the integrand. We shall also prove a second conservation theorem which reduces Dirichlet's multiple integral to a single integral.

DEFINITION 4.4-1 Let $b \in \mathbb{C}_>^{\,k}$, $k \geq 2$, and let $E = E_{k-1}$ be the standard simplex in $\mathbb{R}^{k-1}$. The complex measure μ_b defined on E by

$$d\mu_b(u) = \frac{1}{\mathrm{B}(b)} u_1^{b_1 - 1} \cdots u_{k-1}^{b_{k-1} - 1} (1 - u_1 - \cdots - u_{k-1})^{b_k - 1} \, du_1 \cdots du_{k-1} \tag{1}$$

will be called a Dirichlet measure.

By (4.3-4) the simplex E has unit measure:

$$\mu_b(E) = \int_E d\mu_b(u) = 1. \tag{2}$$

Define $u_k = 1 - u_1 - \cdots - u_{k-1}$ and $\operatorname{Re} b = (\operatorname{Re} b_1, \ldots, \operatorname{Re} b_k)$. The total variation measure $|\mu_b|$ is given by

$$d|\mu_b|(u) = \frac{1}{|\mathrm{B}(b)|} \prod_{i=1}^{k} u_i^{\operatorname{Re} b_i - 1}\, du_1 \cdots du_{k-1} = \frac{\mathrm{B}(\operatorname{Re} b)}{|\mathrm{B}(b)|}\, d\mu_{\operatorname{Re} b}(u), \tag{3}$$

$$|\mu_b|(E) = \frac{\mathrm{B}(\operatorname{Re} b)}{|\mathrm{B}(b)|} < \infty. \tag{4}$$

Because this is finite, an infinite series which converges uniformly on E may be integrated term by term with respect to μ_b.

Let f be a complex-valued function of the nonnegative weights $u_1, \ldots, u_k$ with unit sum. By expressing u_k in terms of the other weights, we obtain a function φ which is defined on E:

$$\varphi(u_1, \ldots, u_{k-1}) = f(u_1, \ldots, u_k), \qquad u_k = 1 - u_1 - \cdots - u_{k-1}. \tag{5}$$

The statement that f is defined and continuous on E means strictly that φ is defined and continuous on E. Since E is compact, f is bounded on E and integrable with respect to μ_b.

THEOREM 4.4-2 Let μ_b be a Dirichlet measure on the standard simplex E in $\mathbb{R}^{k-1}$, $k \geq 2$, and let the function $f(u_1, \ldots, u_k)$, where $\sum_{i=1}^{k} u_i = 1$, be continuous on E. Define

$$g(b_1, \ldots, b_k) = \int_E f(u_1, \ldots, u_k)\, d\mu_b(u). \tag{6}$$

Then any permutation of the subscripts $1, \ldots, k$ applied to $f(u_1, \ldots, u_k)$ induces the same permutation applied to $g(b_1, \ldots, b_k)$.

Proof Substitution of (1) and relabeling of integration variables proves the result for permutations of $1, \ldots, k-1$. Hence it suffices to consider the transposition of 1 and k. In the integral

$$\int_E f(u_k, u_2, \ldots, u_{k-1}, u_1)\, d\mu_b(u) \tag{7}$$

we substitute (1) and change the variables of integration from $u_1, \ldots, u_{k-1}$ to $u_2, \ldots, u_k$, following the same steps which resulted in (4.3-10). By relabeling u_1 and u_k the resulting integral is equal to the right side of (6) with b_k and b_1 interchanged in $d\mu_b(u)$. Hence (7) is equal to $g(b_k, b_2, \ldots, b_{k-1}, b_1)$. □

As an illustration let $m \in \mathbb{N}^k$ and consider the moments of μ_b. By (1),

$$\begin{aligned} u_1^{m_1} \cdots u_k^{m_k}\, d\mu_b(u) &= \frac{\mathrm{B}(b+m)}{\mathrm{B}(b)}\, d\mu_{b+m}(u), \\ \int_E u_1^{m_1} \cdots u_k^{m_k}\, d\mu_b(u) &= \frac{\mathrm{B}(b+m)}{\mathrm{B}(b)} = \frac{(b_1, m_1) \cdots (b_k, m_k)}{(b_1 + \cdots + b_k, m_1 + \cdots + m_k)}. \end{aligned} \tag{8}$$

Any permutation of the u_i in the product $u_1^{m_1} \cdots u_k^{m_k}$ induces the same permutation of the b_i in the last member of the second equation because each is equivalent to the inverse permutation of the m_i.

If $f(u_1, \ldots, u_k)$ is symmetric in a subset of the indices $1, \ldots, k$, it follows from Theorem 4.4-2 that $g(b_1, \ldots, b_k)$ has the same symmetry. In other words, integration with respect to Dirichlet measure conserves permutation symmetry. It also conserves sums in this sense:

THEOREM 4.4-3 Let $p - 1,\ q - 1 \in \mathbb{N}$, and let μ_β and μ_b be Dirichlet measures on E_p and E_{p+q-1}, respectively. Suppose the complex-valued function $f(u_1, \ldots, u_{p+1})$, where $\sum_{i=1}^{p+1} u_i = 1$, is continuous on E_p, and define

$$g(\beta_1, \ldots, \beta_{p+1}) = \int_{E_p} f(u_1, \ldots, u_{p+1})\, d\mu_\beta(u). \tag{9}$$

If $\sum_{i=1}^{p+q} v_i = 1$, then

$$\int_{E_{p+q-1}} f\left(v_1, \ldots, v_p, \sum_{j=1}^{q} v_{p+j}\right) d\mu_b(v) = g\left(b_1, \ldots, b_p, \sum_{j=1}^{q} b_{p+j}\right). \tag{10}$$

Remark According to Theorem 4.4-2 it is immaterial which subscripts occur in the sum in the last argument of f, but the same ones must occur in the last argument of g.

Proof We may assume $q \geq 2$, since (10) reduces to (9) if $q = 1$. Substitute $v_i = u_i$, $i = 1, \ldots, p$, and $v_{p+j} = u_{p+1} w_j$, $j = 1, \ldots, q$, where $u_{p+1} = 1 - u_1 - \cdots - u_p$ and $\sum_{j=1}^{q} w_j = 1$. The Jacobian determinant of the transformation from variables $v_1, \ldots, v_{p+q-1}$ to $u_1, \ldots, u_p, w_1, \ldots, w_{q-1}$ is easily found to be u_{p+1}^{q-1}. In terms of the new variables the region of integration is $E_p \times E_{q-1}$. The integral in (10) becomes

$$\frac{1}{\mathrm{B}(b)} \int_{E_{q-1}} \int_{E_p} f(u_1, \ldots, u_p, u_{p+1}) \left(\prod_{i=1}^{p} u_i^{b_i - 1}\right) u_{p+1}^{\sigma - 1} \prod_{j=1}^{q} w_j^{b_{p+j} - 1}\, du_1 \cdots du_p \cdot dw_1 \cdots dw_{q-1}, \tag{11}$$

where $\sigma = \sum_{j=1}^{q} b_{p+j}$. If the b_i are real and positive and if f is real and nonnegative, then the integrand is nonnegative, and we may evaluate the integral as an iterated integral according to Fubini's theorem (see Appendix B.2). By (4.3-4) and (9) we obtain

$$\begin{aligned}&\frac{1}{\mathrm{B}(b_1, \ldots, b_{p+q})} \mathrm{B}(b_1, \ldots, b_p, \sigma) g(b_1, \ldots, b_p, \sigma) \mathrm{B}(b_{p+1}, \ldots, b_{p+q}) \\ &\quad = g(b_1, \ldots, b_p, \sigma).\end{aligned}$$

In the complex case the absolute value of the integrand is obtained by replacing f by $|f|$ and each b_i by its positive real part. The iterated integral of the absolute value is finite, according to the evaluation just made in the real case. Hence Fubini's theorem justifies the evaluation of (1) as an iterated integral in the complex case also. □

COROLLARY 4.4-4 Let p and q be positive integers, and let μ_b and μ_β be Dirichlet measures on E_{p+q-1} and E_1, respectively. Suppose that the complex-valued function $\varphi(u)$ is continuous for $0 \leq u \leq 1$. Then

$$\int_{E_{p+q-1}} \varphi(v_1 + \cdots + v_p)\, d\mu_b(v) = \int_0^1 \varphi(u)\, d\mu_\beta(u), \tag{12}$$

where $\beta_1 = b_1 + \cdots + b_p$ and $\beta_2 = b_{p+1} + \cdots + b_{p+q}$.

Proof Define f [compare (5)] by $f(u, v) = \varphi(u)$ for $0 \leq u \leq 1$ and $v = 1 - u$. By two applications of Theorem 4.4-3,

$$\begin{aligned}\int_{E_{p+q-1}} & f(v_1 + \cdots + v_p, v_{p+1} + \cdots + v_{p+q})\, d\mu_b(v) \\ &= g(b_1 + \cdots + b_p, b_{p+1} + \cdots + b_{p+q}) \\ &= \int_0^1 f(u, 1-u)\, d\mu_\beta(u).\end{aligned} \tag{13}$$

Expressing f in terms of φ yields (12). □

COROLLARY 4.4-5 Let E_p be the standard simplex in $\mathbb{R}^p$, $p \geq 1$, and let $b \in \mathbb{C}_>{}^p$. Suppose that the complex-valued function $\varphi(u)$ is continuous for $0 \leq u \leq 1$. Then

$$\begin{aligned}\int_{E_p} & \varphi(v_1 + \cdots + v_p) v_1^{b_1 - 1} \cdots v_p^{b_p - 1}\, dv_1 \cdots dv_p \\ &= \mathrm{B}(b_1, \ldots, b_p) \int_0^1 \varphi(u) u^{b_1 + \cdots + b_p - 1}\, du.\end{aligned} \tag{14}$$

Proof Set $q = 1$ and $b_{p+1} = 1$ in Corollary 4.4-4. □

The interest of these two corollaries lies in the reduction of a multiple integral to a single integral. The left side of (14) is often called Dirichlet's multiple integral, although Dirichlet (1839) considered primarily the case $\varphi = 1$; Liouville (1839) was the first to prove (14). Both (12) and (14) will be rewritten in a convenient notation in the next chapter (see Example 5.2-5).

In the proof of the next theorem and subsequent discussions, we shall write

$$[x \mapsto f(x, y)] \quad \text{or} \quad x \mapsto f(x, y)$$

to mean $f(x, y)$ considered as a function of x for fixed y.

THEOREM 4.4-6 For every $b \in \mathbb{C}_>^k$, $k \geq 2$, let μ_b be a Dirichlet measure on the standard simplex E in $\mathbb{R}_{k-1}$. Define $u_k = 1 - u_1 - \cdots - u_{k-1}$ and let the complex-valued function $f(u_1, \ldots, u_k)$ be continuous on E. Define

$$g(b) = \int_E f(u_1, \ldots, u_k)\, d\mu_b(u).$$

Then g is holomorphic on $\mathbb{C}_>^k$.

Proof Write $f(u) = f(u_1, \ldots, u_k)$, denote Lebesgue measure on E by $dm(u) = du_1 \cdots du_{k-1}$, and define $u' = (u_1, \ldots, u_{k-1})$. If ∂E denotes the boundary of E, define

$$p(b, u) = \prod_{i=1}^{k} u_i^{b_i - 1}, \qquad u' \in \operatorname{int}(E),$$

$$p(b, u) = 0, \qquad u' \in \partial E.$$

Then

$$g(b) = \frac{1}{\mathrm{B}(b)} \int_E f(u) p(b, u)\, dm(u).$$

Since $1/\mathrm{B}$ is holomorphic on $\mathbb{C}_>^k$, it suffices to show that the integral is holomorphic on $\mathbb{C}_>^k$. In Theorem B.4 choose $\mu = m$, $T = E$, $Z = \mathbb{C}_>^k$, and verify the three hypotheses of that theorem:

(i) $[b \mapsto f(u)p(b, u)]$ is plainly holomorphic on $\mathbb{C}_>^k$ for every fixed $u' \in E$.

(ii) $[u' \mapsto f(u)p(b, u)]$ is in $L_1(m)$ for every fixed $b \in \mathbb{C}_>^k$, because f is continuous and bounded on the compact set E and p is in $L_1(m)$ by (4.4-4).

(iii) If K is a compact set in $\mathbb{C}_>^k$, there exists $\varepsilon > 0$ such that $b \in K$ implies $\operatorname{Re} b_i > \varepsilon$, $i = 1, \ldots, k$. Let M be a constant such that $|f| \leq M$ on E, and define $g_K(u) = Mp(\beta, u)$, where $\beta = (\varepsilon, \ldots, \varepsilon)$. Then $g_K \in L_1(m)$ and

$$|f(u)p(b, u)| \leq g_K(u)$$

for every $(b, u') \in K \times E$.

If follows from Theorem B.4 that the integral is holomorphic on $\mathbb{C}_>^k$. $\square$

NOTES

The references listed at the end of Chapter 3 also discuss the beta function of two variables. For Dirichlet's multiple integral see Dirichlet (1839), Liouville (1839), Whittaker (1927, p. 258), the detailed treatment by Edwards (1921, Vol. 2, pp. 153–179), and recent extensions by Klamkin (1971).

FORMULARY

$$\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}, \qquad x, y \in U, \tag{4.2-1}$$

$$\mathrm{B}(x, y) = \int_0^1 u^{x-1}(1-u)^{y-1}\,du \qquad x, y \in \mathbb{C}_>, \tag{4.2-2}$$

$$\mathrm{B}(x, y) = 2\int_0^{\pi/2} (\cos\theta)^{2x-1}(\sin\theta)^{2y-1}\,d\theta \qquad x, y \in \mathbb{C}_>, \tag{4.2-3}$$

$$\mathrm{B}(x, y) = \int_0^\infty t^{x-1}(t+1)^{-x-y}\,dt, \qquad x, y \in \mathbb{C}_>, \tag{4.2-4}$$

$$\mathrm{B}(b) = \mathrm{B}(b_1, \ldots, b_k) = \frac{\Gamma(b_1)\cdots\Gamma(b_k)}{\Gamma(b_1 + \cdots + b_k)}, \qquad b \in U^k, \tag{4.3-1}$$

$$\mathrm{B}(b) = \int_E u_1^{b_1-1}\cdots u_{k-1}^{b_{k-1}-1}(1-u_1-\cdots-u_{k-1})^{b_k-1}\,du_1\cdots du_{k-1}, \qquad b \in \mathbb{C}_>{}^k, \tag{4.3-4}$$

$$d\mu_b(u) = \frac{1}{\mathrm{B}(b)} u_1^{b_1-1}\cdots u_{k-1}^{b_{k-1}-1}(1-u_1-\cdots-u_{k-1})^{b_k-1}\,du_1\cdots du_{k-1}, \tag{4.4-1}$$

$$\mu_b(E) = \int_E d\mu_b(u) = 1, \qquad b \in \mathbb{C}_>{}^k, \tag{4.4-2}$$

$$\int_E u_1^{m_1}\cdots u_k^{m_k}\,d\mu_b(u) = \frac{(b_1, m_1)\cdots(b_k, m_k)}{(b_1 + \cdots + b_k, m_1 + \cdots + m_k)}, \qquad m_1, \ldots, m_k \in \mathbb{N}, \quad b \in \mathbb{C}_>{}^k. \tag{4.4-8}$$

EXERCISES

4.2-1 Prove (4.2-2) by using the convolution theorem for Fourier transforms.

4.2-2 Use (4.2-3) to prove

(i) $\Gamma(\frac{1}{2}) = \pi^{1/2}$,

(ii) $\mathrm{B}(z, z) = 2^{1-2z}\mathrm{B}(z, \frac{1}{2})$, $\operatorname{Re} z > 0$.

From (i) and (ii) deduce Legendre's duplication theorem (3.7-2).

4.2-3 If x and y are positive and unequal, show that

$$\mathrm{B}(x, y) > \mathrm{B}\left(\frac{x+y}{2}, \frac{x+y}{2}\right).$$

4.2-4 If $p, q \in \mathbb{C}$ and $\operatorname{Re} q > \operatorname{Re} p > -1$, show that

$$\int_0^\infty (\sinh x)^p(\cosh x)^{-q}\,dx = \frac{1}{2}\mathrm{B}\left(\frac{p+1}{2}, \frac{q-p}{2}\right).$$

4.2-5 Let $I_n = \int_0^1 (1-t^n)^{-1/2}\,dt$, and show that

$$I_3 = \pi^{-1}2^{-4/3}3^{-1/2}[\Gamma(\tfrac{1}{3})]^3, \qquad I_4 = \pi^{-1/2}2^{-5/2}[\Gamma(\tfrac{1}{4})]^2, \qquad I_6 = \pi^{-1}2^{-7/3}[\Gamma(\tfrac{1}{3})]^3.$$

4.2-6 If $0 < \operatorname{Re} a < 1$, show that

$$\int_{-\infty}^\infty \frac{e^{ax}}{e^x + 1}\,dx = \frac{\pi}{\sin \pi a}.$$

4.2-7 Find the area in the xy plane enclosed by the curve $|x|^a + |y|^a = 1$, where $a > 0$. Verify that the answer is correct in the cases $a = 2$, $a = 1$, $a \to \infty$, $a \to 0$.

4.2-8 Bernoulli's lemniscate is a curve in the shape of a figure eight having the equation (in plane polar coordinates) $r^2 = 2a^2 \cos 2\theta$, $a > 0$. Show that its perimeter is $a\mathrm{B}(\frac{1}{4}, \frac{1}{2})$. (Cf. Exercises 8.3-7 and 6.10-3.)

4.2-9 Assume $x, y \in \mathbb{C}$; $x \neq y$; $\beta, \beta' \in \mathbb{C}_>$. Show that

$$\int_y^x (t - y)^{\beta - 1}(x - t)^{\beta' - 1}\, dt = (x - y)^{\beta + \beta' - 1}\mathrm{B}(\beta, \beta').$$

4.2-10 Let $x, \beta, \beta' \in \mathbb{C}_>$ and assume $|\mathrm{ph}(u + x)| < \pi$ for $0 \leq u \leq 1$. Show that

$$\int_0^1 u^{\beta - 1}(1 - u)^{\beta' - 1}(u + x)^{-\beta - \beta'}\, du = \mathrm{B}(\beta, \beta')(1 + x)^{-\beta}x^{-\beta'}.$$

4.2-11 If $x, y \in \mathbb{C}_>$, show that

$$\mathrm{B}(x, y) = \int_0^1 (u^{x - 1} + u^{y - 1})(1 + u)^{-x - y}\, du.$$

4.2-12 If $\mathrm{Re}\, x > |\mathrm{Re}\, y|$, show that

$$\int_0^\infty (\cosh t)^{-2x} \cosh 2yt\, dt = 4^{x - 1}\mathrm{B}(x + y, x - y).$$

4.2-13 If $n \in \mathbb{N}$ and $y \in U$, show that the residue of $[x \mapsto \mathrm{B}(x, y)]$ at $x = -n$ is $(1 - y, n)/n!$. (Cf. Exercise 8.3-5.)

4.2-14 If $x \in U$ and $|\mathrm{ph}\, y| \leq \pi - \varepsilon < \pi$, show that

$$\lim_{y \to \infty} y^x\mathrm{B}(x, y) = \Gamma(x).$$

4.3-1 If $b_1, \ldots, b_k \in \mathbb{C}_>$, show that $|\mathrm{B}(b_1, \ldots, b_k)| \leq \mathrm{B}(\mathrm{Re}\, b_1, \ldots, \mathrm{Re}\, b_k)$. If $b_1 - 1, \ldots, b_k - 1 \in \mathbb{C}_>$, show that $|\mathrm{B}(b_1, \ldots, b_k)| \leq 1/(k - 1)!$.

4.3-2 Assume $b_1, \ldots, b_k$ are positive and not all equal, and define $c = b_1 + \cdots + b_k$. Show that

$$\mathrm{B}(b_1, \ldots, b_k) > \mathrm{B}(c/k, \ldots, c/k).$$

4.3-3 Let $b_1, \ldots, b_k \in U$. For each $i = 1, \ldots, k$, define m_i to be the least nonnegative integer such that $\mathrm{Re}(b_i + m_i) > 0$. Show that the beta function has the integral representation, analogous to (3.2-7) for the gamma function,

$$\mathrm{B}(b) = \frac{(b_1 + \cdots + b_k, m_1 + \cdots + m_k)}{(b_1, m_1) \cdots (b_k, m_k)} \int_E \prod_{i=1}^k u_i^{b_i + m_i - 1}\, du_1 \cdots du_{k-1}.$$

4.3-4 If $\beta \in U$, show that the beta function in k variables, $k \geq 2$, satisfies

$$\mathrm{B}(\beta, \ldots, \beta) = k^{1 - k\beta}\mathrm{B}\left(\beta, \frac{1}{k}\right)\mathrm{B}\left(\beta, \frac{2}{k}\right) \cdots \mathrm{B}\left(\beta, \frac{k - 1}{k}\right).$$

4.3-5 Evaluate the right side of (4.3-4) for $k = 3$ by substituting $u_1 = \sin^2 \theta \cos^2 \varphi$ and $u_2 = \sin^2 \theta \sin^2 \varphi$.

4.3-6 The equation of an ellipsoid in $\mathbb{R}^n$ is $\sum_{i=1}^n x_i^2/a_i^2 = 1$, where $2a_i$ is the length of the ith principal axis. Show that the volume is $\pi^{n/2}a_1 \cdots a_n[\Gamma(\frac{1}{2}n + 1)]^{-1}$.

4.3-7 Show that a sphere in $\mathbb{R}^n$ with radius r has surface area $2\pi^{n/2}[\Gamma(\frac{1}{2}n)]^{-1}r^{n-1}$.

4.3-8 In spherical polar coordinates the axially symmetric volume element in $\mathbb{R}^3$ is

$2\pi r^2 \sin\theta\, dr\, d\theta$. If axial symmetry in $\mathbb{R}^2$ is taken to mean reflection symmetry in the axis $\theta = 0$, the corresponding element in $\mathbb{R}^2$ is $2r\, dr\, d\theta$. Show that the corresponding element in $\mathbb{R}^n$, $n \geq 2$, is

$$2\pi^{(n-1)/2}\left[\Gamma\left(\frac{n-1}{2}\right)\right]^{-1} r^{n-1}(\sin\theta)^{n-2}\, dr\, d\theta.$$

This result will be used in Exercise 5.8-12.

4.3-9 The moment of inertia about the z axis of a body of density $\rho(x, y, z)$ is $\int (x^2 + y^2)\rho(x, y, z)\, dx\, dy\, dz$. Consider a solid ellipsoid of uniform density with semiaxes a, b, c and mass M. Show that its moment of inertia about the semiaxis of length c is $M(a^2 + b^2)/5$.

4.4-1 Show that (4.4-8) implies Vandermonde's theorem (2.3-5).

4.4-2 Let $a \in \mathbb{C}$ and let μ_b be a Dirichlet measure on the unit interval. Prove the integral representations

$$_1F_1(b_1; b_1 + b_2; x) = \int_0^1 e^{ux}\, d\mu_b(u), \qquad x \in \mathbb{C},$$

$$_2F_1(a, b_1; b_1 + b_2; x) = \int_0^1 (1 - ux)^{-a}\, d\mu_b(u), \qquad |x| < 1.$$

The second integral has a meaning unless x is real and $x \geq 1$. It provides the analytic continuation of the hypergeometric function to the x plane cut along the positive real axis from 1 to ∞ (cf. Section 6.8).

4.4-3 Let $a \in \mathbb{C}$ and let μ_b and μ_β be Dirichlet measures on the unit interval. If $|x| < 1$, show that

$$_3F_2\left[\begin{matrix} a, b_1, \beta_1 \\ b_1 + b_2, \beta_1 + \beta_2 \end{matrix}; x\right] = \int_0^1 \int_0^1 (1 - uvx)^{-a}\, d\mu_b(u)\, d\mu_\beta(v).$$

4.4-4 Let ρ be the radius of convergence of the series $_pF_q[(a); (c); x]$, and let μ_b be a Dirichlet measure on the unit interval. Let $(a), b_1$ stand for the array $a_1, \ldots, a_p, b_1$. If $|x| < \rho$, show that

$$\int_0^1 {}_pF_q[(a); (c); ux]\, d\mu_b(u) = {}_{p+1}F_{q+1}[(a), b_1; (c), b_1 + b_2; x],$$

$$\int_0^1 {}_pF_q[(a); (c); 4u(1 - u)x]\, d\mu_b(u) = {}_{p+2}F_{q+2}[(a), b_1, b_2; (c), \tfrac{1}{2}(b_1 + b_2), \tfrac{1}{2}(b_1 + b_2 + 1); x].$$

4.4-5 Let $a, \alpha \in \mathbb{R}_>^k$ and $p \in \mathbb{C}_>^k$, and define $b = (b_1, \ldots, b_k) = (p_1/\alpha_1, \ldots, p_k/\alpha_k)$ and $c = \sum_{i=1}^k b_i$. Let $T \in \mathbb{R}_>$ and let $f\colon [0, T] \to \mathbb{C}$ be continuous. Define $S(T) = \{x \in \mathbb{R}_+^k\colon \sum_{i=1}^k (x_i/a_i)^{\alpha_i} \leq T\}$. Show that

$$\int_{S(T)} f\left[\sum_{i=1}^k (x_i/a_i)^{\alpha_i}\right] \prod_{i=1}^k x_i^{p_i - 1}\, dx_1 \cdots dx_k = \prod_{i=1}^k (\alpha_i^{-1} a_i^{p_i}) \mathrm{B}(b) \int_0^T f(t) t^{c-1}\, dt.$$

As $T \to \infty$ the equation still holds if the limits of the integrals exist; note that $S(\infty) = \mathbb{R}_+^k$.

4.4-6 Let $f(t)$ be real, nonnegative, continuous for $0 \leq t \leq T$, and zero for $t > T$. If $T = \infty$ assume further that $t^{3/2}f(t)$ is integrable on $\mathbb{R}_+$. Let $a, b, c \in \mathbb{R}_>$ and consider a three-dimensional distribution of mass with density $\rho(x, y, z) = f[(x/a)^2 + (y/b)^2 + (z/c)^2]$, $(x, y, z) \in \mathbb{R}^3$. Show that the total mass M and the moment of inertia I_z about the z axis (see Exercise 4.3-9) are given by

$$M = 2\pi abc \int_0^T t^{1/2} f(t)\, dt, \qquad I_z = \frac{2\pi abc}{3}(a^2 + b^2) \int_0^T t^{3/2} f(t)\, dt.$$

4.4-7 Let $p, q, r, s \in \mathbb{C}_>$. Show that

$$\int_{\mathbb{R}_+^4} x^{p-1}y^{q-1}z^{r-1}w^{s-1}e^{-(x+y+z+w)^2}\,dx\,dy\,dz\,dw = \frac{1}{2}\,\mathrm{B}(p, q, r, s)\Gamma\left(\frac{p+q+r+s}{2}\right).$$

4.4-8 Assume $\operatorname{Re} b_i \geq \frac{1}{2}$ for $i = 1, \ldots, k$. Show that

$$|\mu_b|(E) \leq \exp\left(\tfrac{1}{2}\pi \sum_{i=1}^{k} |\operatorname{Im} b_i|\right).$$